



COMPETENCY STANDARD

FOR

INDUSTRIAL AUTOMATION

(LIGHT ENGINEERING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Industrial Automation serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored for the Light Engineering sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical for Industrial Automation. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-IA-01-G	Apply Occupational Health and Safety (OHS) Practices in Workplace	1. Identify OHS policies and procedures 2. Practice personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-LE-IA-02-G	Operate in a Self-Directed Team	1. Identify team goals and work processes 2. Communicate and cooperate with team members. 3. Work as a team member. 4. Solve problems as a team member	10
Total Hour			20

Sector Specific Competencies (10 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-IA-01-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hour			10

Occupation Specific Competencies (270 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-IA-01-O	Execute Electrical Sequence Circuit and Control Using Relay Logic	1. Interpret Relay Logic Operation 2. Perform Mechanical Switching Operation 3. Execute Timer and Counter Application 4. Construct Latch, DOL And Star Delta 5. Perform Automatic Forward Reverse Operation	60
SICIP-LE-IA-02-O	Implement PLC Installation and Wiring with Configuration	1. Prepare for Works 2. Set up PLC Hardware Configurations with Software Tool 3. Establish Communication Between PLC And PC	15
SICIP-LE-IA-03-O	Carry Out Ladder Diagram in PLC Programming	1. Prepare for Ladder Diagram 2. Perform PLC programming using ladder diagram 3. Prepare PLC for operation	35
SICIP-LE-IA-04-O	Integrate Industrial Sensors and Actuators	1. Identify types of industrial sensors and classifications 2. Identify types of industrial actuators and classifications 3. Select Sensors Based on Application and Material Properties 4. Execute Sensors and Actuators for Industrial Application	40
SICIP-LE-IA-05-O	Perform Process Visualization (HMI)	1. Configure Human Machine Interface (HMI) Panel Integration 2. Perform Process Design in HMI 3. Carry out interface between PLC and HMI 4. Visualize the Process in HMI	20
SICIP-LE-IA-06-O	Operate Pneumatic and Hydraulic System Using PLC	1. Identify Pneumatic and Hydraulic System Components 2. Install and Configure PLC	30

		3. Develop PLC Programs for Pneumatic Control System 4. Develop PLC Programs Hydraulic Control System 5. Test and run PLC-Controlled Pneumatic and Hydraulic Systems	
SICIP-LE-IA-07-O	Control Variable Frequency Drive (VFD) using PLC	1. Interpret Variable Frequency Drive (VFD) 2. Install and configure Variable Frequency Drive (VFD) 3. Operate Variable Frequency Drive (VFD) in Different Modes	15
SICIP-LE-IA-08-O	Design and Develop PLC Based Control System	1. Analyze Control System Requirements 2. Design PLC Control System Architecture 3. Develop PLC Program 4. Simulate the PLC Program 5. Install and Configure PLC Hardware 6. Commission and Debug Control System	55
Total Hour			0

COMPETENCY STANDARD: INDUSTRIAL AUTOMATION

The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-IA-01-G
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Practice personal health and safety procedures	2.1 OHS policies and procedures are followed and practiced. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal health, hygiene and safety procedures are practiced.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. OHS policies	1.1 International/ Local OHS requirements 1.2 Fire Safety Rules and Regulations

	1.3 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gloves 2.3 Safety shoes 2.4 Helmet 2.5 Face mask 2.6 Goggles and safety glasses 2.7 Ear plugs 2.8 Sun block 2.9 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards 3.2 Biological hazards 3.3 Physical Hazards 3.3.1 Machine hazards 3.3.2 Materials hazards 3.3.3 Tools and Equipment hazards
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Applying CPR (Cardiopulmonary Resuscitation) 5.5 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures 1.2 Work safety procedures 1.3 Emergency procedures 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Explosion response 1.3.4 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects. 1.5 PPE types and uses 1.6 Personal hygiene practices 1.7 OHS awareness 1.8 Malfunctions and resolutions
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Applying personal work safety practices in the workplace 2.3 Identifying, assessing and controlling hazards and risks 2.4 Reporting incidents arising from hazards and risks 2.5 Responding to emergency procedures 2.6 Maintaining physical well-being in the workplace 2.7 Using firefighting accessories and fire extinguishers

	2.8 Implementing emergency response plans and procedures 2.9 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits 4.6 Learning manual

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 described emergency procedures 1.2 followed OHS policies and procedures 1.3 selected and used personal protective equipment (PPE) 1.4 practiced personal health and safety procedures 1.5 identified, assessed and controlled hazards and risks 1.6 reported incidents arising from hazards and risks to the authority 1.7 implemented plans and procedures to respond emergency 1.8 applied basic first aid procedure
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IN A SELF-DIRECTED TEAM	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-IA-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood, valued and practiced. 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practices. 3.3 Team member's support to other members is appreciated and valued. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables

Variable	Range
	May Include but not limited to:

1. Forms of communication	1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OHS information 1.9 Signs
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Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective interpersonal skills to interact with team members 1.5 Effective formal and informal forms of communication 1.6 Value of diversity in team functioning 1.7 Correct use of workplace terminology 1.8 Team's duties, responsibilities, authorities, objectives and task requirements 1.9 Methods of identifying current and potential problems faced by a team 1.10 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Identifying relationships within team and with other workers 2.4 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives 2.5 Using formal and informal forms of communication 2.6 Understanding and valuing views and opinions of other team members 2.7 Performing tasks in accordance with organizational and team requirements, and workplace procedures and practice 2.8 Appreciating and valuing team member's support to other members of the team 2.9 Identifying current and potential problems faced by the team 2.10 Identifying solutions to the problem 2.11 Solving problems effectively and evaluating the outcome of the implemented solution

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Work book 4.3 Learning & operational manuals 4.4 Workplace tools/equipment

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 described team's duties, responsibilities, authorities, objectives and task requirements 1.2 identified team goals and work processes 1.3 communicated and cooperated with team members 1.4 used effective interpersonal skills to interact with team members 1.5 worked as a team member 1.6 solved problems as a team member
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

The Sector Specific Competencies

Unit of Competency: APPLY GREEN PRACTICES	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-IA-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace, and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices</u> are explained. 1.2 <u>Sources of environmental impacts</u> during light engineering activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 <u>Ways of mitigating environmental impacts</u> in light engineering sector are explained.
2. Minimize resource use in the workplace	2.1 Water, energy and raw material consumption are documented. 2.2 <u>Recyclable and non-recyclable items</u> are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of wastes</u> are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce wastes in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair 1.3 Use of sustainable materials with low environmental

	impact 1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of environmental impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical's leakage 2.3 Noise pollution 2.4 Waste generation 2.4.1 Waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment 3.2 Adopting renewable energy sources 3.3 Implementing site protection measures 3.4 Using reusable materials 3.5 Choosing sustainable materials 3.6 Using noise-reducing equipment and scheduling work appropriately 3.7 Optimizing logistics and delivery
4. Recyclable and non-recyclable items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different types of wastes	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Electrical system waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste 5.8 Chemical West 5.9 Fuel/oil west

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in light engineering sector 1.2 Ways of mitigating environmental impacts in light engineering sector 1.3 Key terms and symbols related to environmental sustainability in light engineering drawings 1.4 Sources of environmental impacts in light engineering 1.5 Methods to minimize resource consumption (water, energy, raw materials) 1.6 Waste management practices 1.7 Sustainable alternatives to fossil-based energy resources 1.8 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	2.1 Identifying sources of environmental impacts during light engineering activities 2.2 Identifying recyclable and non-recyclable items 2.3 Exploring and applying sustainable alternatives to fossil-based energy resources 2.4 Applying methods to minimize resource use in the workplace 2.5 Implementing waste management practices 2.6 Applying procedures for safely disposing of hazardous materials 2.7 Identifying different types of wastes 2.8 Documenting water, energy and material consumption in the workplace 2.9 Performing green practices in personal and professional activities
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of light engineering hand tools and power tools 4.3 Sustainable, re-cyclable, reusable materials 4.4 Work books 4.5 Operation and maintenance manuals 4.6 Waste segregation bins 4.7 Learning manual 4.8 Energy-efficient equipment

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 explained the principles of green practices in light engineering 1.2 identified sources of environmental impacts 1.3 implemented procedures to minimize resource use (water, energy, raw materials) 1.4 explained ways of mitigating environmental impacts in light engineering sector 1.5 disposed of hazardous waste in compliance with safety and environmental standards 1.6 practiced green habits to reduce waste in both personal and professional life 1.7 applied methods to minimize resource use in the workplace
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: EXECUTE ELECTRICAL SEQUENCE CIRCUIT AND CONTROL USING RELAY LOGIC	Nominal Duration: 60 hrs.	Unit Code: SICIP-LE-IA-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to execute electrical sequence circuit and control using relay logic. It specifically includes the tasks of interpreting relay logic operation, performing mechanical switching operation, executing timer and counter application, constructing latch, DOL and star delta, and performing automatic forward reverse operation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret Relay Logic Operation	1.1 Fundamental concepts of relay logic are identified and explained according to industrial automation principles. 1.2 <u>Components of relay logic circuits</u> are identified. 1.3 Relay logic components' functions are described. 1.4 Operation of relay logic circuits is interpreted. 1.5 Common relay <u>logic symbols</u> and notations are identified. 1.6 Application of relay logic in industrial automation processes is described.
2. Perform Mechanical Switching Operation	2.1 Mechanical switching devices are identified and described. 2.2 Mechanical switching operation procedures are followed. 2.3 The <u>contact status</u> is checked using multimeter. 2.4 <u>Boolean logic</u> operation using relay is executed. 2.5 Switching devices are operated as per the control system design. 2.6 Mechanical switches are inspected and tested for proper functioning before and after operation.
3. Execute Timer and Counter Application	3.1 <u>Timer</u> and <u>counter</u> applications are selected based on control system needs. 3.2 Timer and counter settings are configured according to application requirements. 3.3 Timer and counter operations are executed following standard procedures. 3.4 Timer and counter outputs are monitored and verified. 3.5 Faults or deviations in timer and counter performance are identified.
4. Construct Latch, DOL	4.1 Circuit diagrams for latch, DOL, and star-delta

and Star Delta	configurations are interpreted. 4.2 Latch circuits are constructed by connecting relays, contacts, and coils as per standard wiring diagrams. 4.3 DOL starter circuits are assembled ensuring proper connections for motor start and stop functions. 4.4 Star-delta starter circuits are constructed following established wiring procedures. 4.5 Circuits are tested for correct operation.
5. Perform Automatic Forward Reverse Operation	5.1 Control circuit for forward and reverse operation is interpreted. 5.2 Automatic forward and reverse operation are executed using timer and counter. 5.3 Operation is tested to verify smooth and correct transition between forward and reverse motion. 5.4 Faults or abnormalities during operation are identified and reported for corrective actions.

Range of Variables

Variable	Range May include but not limited to:
1. Components of relay logic circuits	1.1 Switches 1.2 Relay 1.3 Magnetic Contactor 1.4 Timer 1.5 Counter 1.6 LED (light emitting diode) 1.7 Motor 1.8 Sensors
2. Logic symbols	2.1 Normally Open (NO) 2.2 Normally Close (NC) 2.3 Timer 2.4 Counter 2.5 Relay Coil
3. Contact status	3.1 Normally Open (NO) 3.2 Normally Close (NC)
4. Boolean logic	4.1 AND 4.2 OR 4.3 NAND 4.4 NOR 4.5 XOR 4.6 XNOR
5. Timer	5.1 On-delay timer 5.2 Off-delay timer
6. Counter	6.1 Up counter 6.2 Down counter

Curricular Content Guide

1. Underpinning knowledge	1.1 Fundamental concepts of relay logic 1.2 Relay logic components' functions 1.3 Operation of relay logic circuits 1.4 Mechanical switching operation procedures 1.5 Familiarity with timer and counter application 1.6 Circuit diagrams for latch, DOL and star-delta configurations 1.7 Control circuit for forward and reverse operation
2. Underpinning skills	2.1 Identifying Components of relay logic circuits 2.2 Identifying common relay logic symbols and notations 2.3 Identifying mechanical switching devices 2.4 Executing Boolean logic operation using relay 2.5 Inspecting and testing mechanical switches for proper functioning before and after operation 2.6 Configuring timer and counter settings according to application requirements 2.7 Executing timer and counter operations following standard procedures 2.8 Constructing latch, DOL starter and star-delta starter circuits 2.9 Executing automatic forward and reverse operation using timer and counter
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 Relays, timers, counters, wirings 4.4 Simulation/testing equipment 4.5 Drawings and sketches

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 explained mechanical switching operation procedures 1.2 interpreted timer and counter application 1.3 interpreted circuit diagrams for latch, DOL and star-delta configurations 1.4 explained control circuit for forward and reverse operation 1.5 executed Boolean logic operation using relay
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	1.6 executed timer and counter operations following standard procedures 1.7 constructed latch, DOL and star-delta starter circuits 1.8 executed automatic forward and reverse operation
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IMPLEMENT PLC INSTALLATION AND WIRING WITH CONFIGURATION	Nominal Duration: 15 hrs.	Unit Code: SICIP-LE-IA-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to implement PLC installation and wiring with configuration. It specifically includes the tasks of preparing for works, setting up PLC hardware configurations with software tool and establishing communication between PLC And PC.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for works	1.1 Safe working practices are followed and <u>Personal Protective Equipment (PPE)</u> is worn as per workplace requirement. 1.2 Necessary tools and equipment are selected and prepared for Installation as job requirements. 1.3 Materials and <u>PLC components</u> are identified required for Installation and commissioning of PLC.
2. Setup PLC hardware configurations with software tool	2.1. Typical PLC components and the control devices are installed as per job requirement. 2.2. Sensors and switches are connected to a particular input and output of PLC. 2.3. <u>Cable connections</u> between the PLC and the <u>programming devices</u> are connected as per work instruction. 2.4. Connection is checked as per required specification. 2.5. <u>Protective devices</u> are set as per the user manual. 2.6. Required <u>software tool</u> is installed. 2.7. Selected hardware is configured using PLC software tools.
3. Establish communication between PLC and PC	3.1 Hardware devices are checked in accordance with user manual. 3.2 Address of PLC and PC are configured as per job requirement. 3.3 Communication between PLC and PC using <u>communication protocols</u> is established. 3.4 Communication between PLC & PC is verified.

Range of Variables

Variable	Range
	May include but not limited to:
1. Personal Protective Equipment (PPE)	1.1 Apron 1.2 Safety helmet

	1.3 Insulated Gloves 1.4 Insulated shoe
2. PLC components	2.1 Power supply unit/module 2.2 Input/output module 2.3 Mounting accessories 2.4 CPU unit 2.5 Communication module 2.6 Programming device 2.7 Front connector
3. Cable connections	3.1 RS 232 3.2 USB 3.3 Ethernet 3.4 RS485 3.5 RS422 3.6 Profibus
4. Programming devices	4.1 Computer with programming software 4.2 Laptop with programming software
5. Protective devices	5.1 Fuse 5.2 Miniature Circuit Breaker (MCB)
6. Software tool	6.1 TIA Portal 6.2 GX Works 6.3 RSLOGIX 6.4 LSXG5000 6.5 WPLSoft
7. Communication protocols	7.1 Profinet 7.2 USB 7.3 Modbus

Curricular Content Guide

1. Underpinning knowledge	1.1 Safe working practices and Personal Protective Equipment (PPE) 1.2 Awareness of PLC components, cable connections, programming devices, protective devices, software tools and communications protocols
2. Underpinning skills	2.1 Identifying materials and PLC components required for Installation and commissioning of PLC 2.2 Installing typical PLC components and the control devices as per job requirement 2.3 Making cable connections between the PLC and the programming devices as per work instruction 2.4 Installing required software tool 2.5 Configuring selected hardware using PLC software tools 2.6 Configuring address of PLC and PC as per job requirement 2.7 Establishing communication between PLC and PC using communication protocols
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities

	3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Drawings, job specifications and work instructions 4.4 PLC programming software (matching PLC brand and model) 4.5 Networking equipment (switches, routers, industrial ethernet) 4.6 Programming devices, protective devices, software tools and communications protocols

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 described PLC components, cable connections, programming devices, protective devices, software tools and communications protocols 1.2 installed typical PLC components and control devices as per job requirement 1.3 made cable connections between PLC and programming devices as per work instruction 1.4 installed required software tool 1.5 configured selected hardware using PLC software tools 1.6 established communication between PLC and PC using communication protocols
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT LADDER DIAGRAM IN PLC PROGRAMMING	Nominal Duration: 35 hrs.	Unit Code: SICIP-LE-IA-03-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to carry out ladder diagram in PLC programming. It specifically includes the tasks of preparing for ladder diagram, performing PLC programming using ladder diagram and preparing PLC for operation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for ladder diagram	1.1 Safe work practices are followed and <u>Personal Protective Equipment (PPE)</u> are worn as required. 1.2 Necessary <u>software and hardware</u> are selected as per job requirements. 1.3 Table chart for sub-program, function, function block and data block are prepared. 1.4 <u>Programming components and ladder logic symbols</u> are identified required for ladder diagram of PLC.
2. Perform PLC programming using ladder diagram	2.1. Basic ladder logic symbols and components are identified according to instruction manual. 2.2. Ladder diagram is constructed by arranging symbols logically to represent the desired control sequence. 2.3. Input and output devices are mapped within the ladder diagram. 2.4. Ladder diagram is checked for logical consistency and observed through simulation. 2.5. Modifications and corrections are made to the ladder diagram based on testing results.
3. Prepare PLC for operation	3.1 Programming devices are collected. 3.2 PLC program required by specific application is developed. 3.3 PLC Program is imported from memory card if required. 3.4 PLC Program is compiled with diagnosis. 3.5 Programs are transferred to PLC and set the PLC in Run Mode. 3.6 Optimization of software is done to ensure the program after the first test run. 3.7 Final circuit testing and optimization is done to ensure no errors occur by the entire system.

Range of Variables

Variable	Range
	May include but not limited to:
1. Personal Protective Equipment (PPE)	1.1 Apron / overall 1.2 Goggles 1.3 Insulated Gloves 1.4 Insulated shoe
2. Software and hardware	2.1. Software 2.1.1 TIA Portal 2.1.2 GX Works 2.1.3 RSLOGIX 2.1.4 LSXG5000 2.1.5 WPLSoft 2.2. Hardware 2.1.1 PLC CPU 2.1.2 Input module 2.1.3 Output module 2.1.4 Communication module 2.1.5 Communication cable 2.1.6 Programming cable
3. Programming components and ladder logic symbols	3.1. Make contact 3.2. Break contact 3.3. Analog contact 3.4. Relay coil 3.5. Inverted output 3.6. Analog output 3.7. All types of timers 3.8. Counter 3.9. Comparator 3.10. Mathematical functions

Curricular Content Guide

1. Underpinning knowledge	1.1 Safe work practices and Personal Protective Equipment (PPE) 1.2 Awareness of necessary software and hardware 1.3 Basic ladder logic symbols and components
2. Underpinning skills	2.1 Selecting necessary software and hardware as per job requirements 2.2 Identifying programming components and ladder logic symbols required for ladder diagram of PLC 2.3 Constructing ladder diagram by arranging symbols logically to represent the desired control sequence 2.4 Checking and observing (through simulation) the ladder diagram for logical consistency 2.5 Developing PLC program required by specific application 2.6 Compiling PLC program with diagnosis

	2.7 Transferring programs to PLC and setting the PLC in Run Mode 2.8 Performing final circuit testing and optimization to ensure no occurrence of any error by the entire system
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 PLC programming software 4.4 Drawings, specifications and work instructions 4.5 Learning manual 4.6 PLC unit, control panel 4.7 Diagnostic tools

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 interpreted basic ladder logic symbols and components 1.2 selected necessary software and hardware as per job requirement 1.3 constructed ladder diagram by arranging symbols logically to represent the desired control sequence 1.4 developed PLC program required by specific application 1.5 transferred programs to PLC and set the PLC in Run Mode
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INTEGRATE INDUSTRIAL SENSORS AND ACTUATORS	Nominal Duration: 40 hrs.	Unit Code: SICIP-LE-IA-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to integrate industrial sensors and actuators. It specifically includes the tasks of identifying types of industrial sensors and classifications, identifying types of industrial actuators and classifications, selecting sensors based on application and material properties, and executing sensors and actuators for industrial application.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify types of industrial sensors and classifications	1.1 Types of <u>industrial sensors</u> are interpreted. 1.2 Classifications of industrial sensors are identified and described. 1.3 Function and role of sensors are explained in relation to industrial applications. 1.4 Applications of industrial sensors are interpreted.
2. Identify types of industrial actuators and classifications	2.1 Types of <u>actuators</u> are interpreted. 2.2 Classifications of industrial actuators are identified and described. 2.3 Function and role of actuators are explained in relation to industrial applications. 2.4 Applications of industrial actuators are interpreted.
3. Select sensors based on application and material properties	3.1 Requirements of the application are analyzed to determine appropriate sensor types. 3.2 Relevant material properties and surface characteristics are identified. 3.3 <u>Sensor specifications</u> are reviewed and matched with application needs. 3.4 <u>Sensor technologies</u> are selected based on material properties and operational conditions. 3.5 Installation and operating environment constraints are evaluated to ensure sensor compatibility and reliability.
4. Execute sensors and actuators for industrial application	4.1 Sensors and actuators are selected and collected according to system specifications. 4.2 Installation sites for sensors and actuators are prepared and inspected to ensure compatibility and safety. 4.3 Sensors and actuators are installed and wired following manufacturer guidelines. 4.4 PLC programming is carried out as per job requirement. 4.5 Functional testing is conducted to verify correct sensor detection and actuator response. 4.6 Faults or malfunctions are identified, diagnosed, and rectified according to troubleshooting procedures.

Range of Variables

Variable	Range May include but not limited to:
1. Industrial sensors	1.1 Proximity 1.2 Limit Switch 1.3 Pressure 1.4 Temperature
2. Actuators	2.1 Linear 2.2 Rotary
3. Sensor specifications	3.1 Sensitivity 3.2 Range 3.3 Accuracy 3.4 Environmental suitability
4. Sensor technologies	4.1 Inductive 4.2 Capacitive 4.3 Ultrasonic 4.4 Photoelectric

Curricular Content Guide

1. Underpinning knowledge	1.1 Types and classifications of industrial sensors 1.2 Function and role of sensors in relation to industrial applications 1.3 Types and classifications of industrial actuators 1.4 Function and role of actuators in relation to industrial applications
2. Underpinning skills	2.1 Identifying classifications of industrial sensors 2.2 Identifying classifications of industrial actuators 2.3 Identifying relevant material properties and surface characteristics 2.4 Reviewing and matching sensor specifications with application needs 2.5 Selecting sensor technologies based on material properties and operational conditions 2.6 Installing and wiring sensors and actuators following manufacturer guidelines 2.7 Carrying out PLC programming as per job requirement. 2.8 Executing sensors and actuators for application in industry 2.9 Identifying, diagnosing and rectifying faults or malfunctions according to troubleshooting procedures
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace

	3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource implications	The Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Learning manual 4.4 Software, hardware 4.5 System compatibility, protocols 4.6 Drawings, Job specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 described types and classifications of industrial sensors and actuators 1.2 explained functions and roles of sensors and actuators in relation to industrial applications 1.3 reviewed and matched sensor specifications with application needs 1.4 selected sensor technologies based on material properties and operational conditions 1.5 installed and wired sensors and actuators following manufacturer guidelines 1.6 executed sensors and actuators for application in industry
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM PROCESS VISUALIZATION (HMI)	Nominal Duration: 20 hrs.	Unit Code: SICIP-LE-IA-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to perform process visualization (HMI). It specifically includes the tasks of configuring human machine interface (HMI) panel integration, performing process design in HMI, carrying out interface between PLC and HMI and visualizing the process in HMI.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Configure human machine interface (HMI) panel integration	1.1 Concept and functions of Human Machine Interface (HMI) are interpreted and understood according to system requirements. 1.2 Different <u>HMI communication protocols</u> are established and configured. 1.3 Various HMI screens are created, selected, and programmed with soft buttons. 1.4 <u>Basic objects</u> and <u>elements</u> are created and configured. 1.5 Graphic objects are imported from the software library and integrated. 1.6 Analogue data tags are configured to enable real-time monitoring.
2. Perform process design in HMI	2.1 Process requirements and specifications are reviewed and interpreted. 2.2 Relevant process control parameters are identified into the HMI design. 2.3 Appropriate interactive elements are designed and configured to control and monitor the process. 2.4 Alarm and notification systems are integrated into the HMI. 2.5 Process design is simulated and tested within the HMI.
3. Carry out interface between PLC and HMI	3.1 Communication between the PLC and HMI is established using appropriate communication protocols. 3.2 Data exchange between the PLC and HMI is configured and verified for accuracy and reliability. 3.3 Communication settings are set according to system requirements. 3.4 Network connections and wiring are inspected and confirmed to ensure proper interface operation. 3.5 Troubleshooting procedures are applied to identify and resolve communication issues between PLC and HMI.

	3.6 Interface performance is monitored.
4. Visualize the process in HMI	4.1 The process flow and operational parameters are interpreted and analyzed for visualization purposes. 4.2 Appropriate graphical elements and indicators are selected and configured to represent process variables accurately. 4.3 Process data is integrated and displayed in real-time on HMI screens using charts, gauges, and trend graphs. 4.4 Interactive components and alarms are incorporated to facilitate process control and monitoring. 4.5 The HMI visualization is tested and validated to confirm accuracy and responsiveness.

Range of Variables

Variable	Range May include but not limited to:
1. HMI communication protocols	1.1 Modbus 1.1.1 RS232 1.1.2 RS485 1.2 Profibus 1.3 TCP IP/Ethernet 1.4 ProfNet
2. Basic objects	2.1 Line 2.2 Ellipse 2.3 Circle 2.4 Rectangle 2.5 Text field 2.6 Graphic view
3. Elements	3.1 I/O field 3.2 Symbolic I/O field 3.3 Graphic I/O field 3.4 Date/Time field 3.5 Bar 3.6 Switch 3.7 Button

Curricular Content Guide

1. Underpinning knowledge	1.1 Concept and functions of Human Machine Interface (HMI) 1.2 Process requirements and specifications 1.3 Process flow and operational parameters (for visualization purposes)
2. Underpinning skills	2.1 Establishing and configuring different HMI communication protocols 2.2 Creating and configuring Basic objects and elements 2.3 Identifying relevant process control parameters into the HMI design 2.4 Designing and configuring appropriate interactive elements to control and monitor the process

	2.5 Establishing communication between PLC and HMI using appropriate communication protocols 2.6 Configuring and verifying data exchange between PLC and HMI for accuracy and reliability 2.7 Selecting and configuring appropriate graphical elements and indicators to represent process variables accurately 2.8 Integrating and displaying process data in real-time on HMI screens using charts, gauges and trend graphs
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Learning manual 4.4 Software, hardware 4.5 Drawings, specifications and work instructions 4.6 Communication protocols 4.7 HMI panels (touchscreens, monitors, etc.)

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 interpreted process requirements and specifications 1.2 explained process flow and operational parameters (for visualization purposes) 1.3 established and configured different HMI communication protocols 1.4 identified relevant process control parameters into the HMI design 1.5 established communication between PLC and HMI using appropriate communication protocols 1.6 selected and configured appropriate graphical elements and indicators to represent process variables 1.7 integrated and displayed process data in real-time on HMI screens using charts, gauges and trend graphs
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)

3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: OPERATE PNEUMATIC AND HYDRAULIC CONTROL SYSTEM USING PLC	Nominal Duration: 30 hrs.	Unit Code: SICIP-LE-IA-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to operate pneumatic and hydraulic control system using PLC. It specifically includes the tasks of identifying pneumatic and hydraulic system components, installing and configuring PLC, developing PLC programs for pneumatic control system, developing PLC programs for hydraulic control system and testing, and running PLC controlled pneumatic and hydraulic systems.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Pneumatic and Hydraulic System Components	1.1 Pneumatic system and Hydraulic system are interpreted and described. 1.2 <u>Components for pneumatic and hydraulic control system</u> are recognized and classified. 1.3 The operating principles of pneumatic and hydraulic components are interpreted. 1.4 Safety features associated with pneumatic and hydraulic systems are identified and explained.
2. Install and Configure PLC	2.1 Installation site and equipment for PLC integration with pneumatic and hydraulic systems are inspected and prepared. 2.2 PLC hardware components are installed and connected. 2.3 Wiring between PLC and pneumatic and hydraulic components are carried out. 2.4 PLC software is installed and configured to control pneumatic and hydraulic system operations.
3. Develop PLC Program for Pneumatic Control System	3.1 <u>Valve sequencing</u> is performed using PLC. 3.2 A two-step control system is executed to regulate and maintain pressure within specified limits. 3.3 Three-step control system is executed to provide enhanced pressure control as per system requirements.
4. Develop PLC Program for Hydraulic Control System	4.1 Valve sequencing is performed using PLC according to system requirements. 4.2 Two-step control system is executed to regulate pressure within specified parameters. 4.3 Three-step control system is executed to provide advanced pressure control.
5. Test and run PLC-Controlled Pneumatic and Hydraulic Systems	5.1 Control circuit for controlling pneumatic and hydraulic system is tested. 5.2 Operation of the systems is conducted according to testing protocols.

	5.3 Faults and irregularities during operation are identified, diagnosed, and resolved.
	5.4 System responses to control commands are monitored and recorded.
	5.5 Test results and fault resolution actions are documented.

Range of Variables

Variable	Range
	May include but not limited to:
1. Components for pneumatic and hydraulic control system	1.1 Valves 1.2 Actuators 1.3 Pumps 1.4 Cylinders 1.5 Filters 1.6 Accumulators
2. Valve sequencing	2.1 Double cylinder operation 2.1.1 A+ B+: A- B- 2.1.2 A+B+: B- A- 2.1.3 A+A-: B+ B- 2.2 Lifting device 2.3 Traffic control

Curricular Content Guide

1. Underpinning knowledge	1.1 Pneumatic and Hydraulic control systems 1.2 Components for pneumatic and hydraulic control systems, and their operating principles
2. Underpinning skills	2.1 Identifying safety features associated with pneumatic and hydraulic systems 2.2 Installing and connecting PLC hardware components 2.3 Carrying out wiring between PLC and pneumatic & hydraulic components 2.4 Installing and configuring PLC software to control pneumatic and hydraulic system operations 2.5 Performing valve sequencing using PLC according to system requirements (for pneumatic control system) 2.6 Performing valve sequencing using PLC according to system requirements (for hydraulic control system) 2.7 Executing two-step & three-step control systems to regulate and maintain pressure within specified limits (for pneumatic control system) 2.8 Executing two-step & three-step control systems to regulate and maintain pressure within specified limits (for hydraulic control system) 2.9 Testing and running PLC controlled pneumatic and hydraulic systems 2.10 Identifying, diagnosing and resolving faults and irregularities during operation
3. Underpinning attitudes	3.1 Commitment to occupational health and safety

	3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 PLC hardware & software 4.4 Learning manual 4.5 Drawings, specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 interpreted components for pneumatic and hydraulic control systems and their operating principles 1.2 carried out wiring between installed PLC hardware and pneumatic & hydraulic components 1.3 installed and configured PLC software to control pneumatic and hydraulic system operations 1.4 executed two-step & three-step control systems to regulate and maintain pressure within specified limits (for pneumatic control system) 1.5 executed two-step & three-step control systems to regulate and maintain pressure within specified limits (for hydraulic control system) 1.6 tested and run PLC controlled pneumatic and hydraulic systems
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CONTROL VARIABLE FREQUENCY DRIVE (VFD) USING PLC	Nominal Duration: 15 hrs.	Unit Code: SICIP-LE-IA-07-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to control variable frequency drive (VFD). It specifically includes the tasks of interpreting variable frequency drive (VFD), installing and configuring variable frequency drive (VFD) and operating variable frequency drive (VFD) in different modes.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret Variable Frequency Drive (VFD)	1.1 Functions of Variable Frequency Drive (VFD) are interpreted. 1.2 <u>Key components of VFD</u> are explained and described. 1.3 Common applications of VFDs in industrial automation are identified. 1.4 Specifications of VFD is interpreted.
2. Install and configure Variable Frequency Drive (VFD)	2.1 Installation site for the VFD is inspected. 2.2 VFD unit is installed securely according to manufacturer guidelines. 2.3 Power cables are connected to the VFD following wiring diagrams. 2.4 Motor is connected to the VFD, ensuring proper wiring and compatibility. 2.5 Controlling circuit for the VFD is installed and integrated with the control system. 2.6 <u>Motor parameters</u> are set accurately in the VFD. 2.7 <u>Controlling parameters</u> are configured according to application requirements.
3. Operate Variable Frequency Drive (VFD) in different mode	3.1 Motor speed control is performed using local or keypad control mode as per operational procedures. 3.2 Motor control is executed using terminal control mode or remote-control mode. 3.3 Motor direction control is applied to enable both forward and reverse operations safely. 3.4 VFD alarms and fault indicators are interpreted accurately to diagnose system status. 3.5 Appropriate responses to alarms are implemented following troubleshooting guidelines. 3.6 Operational parameters are monitored and adjusted to maintain desired motor performance.

Range of Variables

Variable	Range May include but not limited to:
1. Key components of VFD	1.1 Inverters 1.2 Rectifiers 1.3 Control units
2. Motor parameters	2.1 Rated voltage 2.2 Rated current 2.3 Frequency 2.4 Number of poles 2.5 Efficiency 2.6 Rated RPM 2.7 Power factor (PF) 2.8 Insulation type
3. Controlling parameters	3.1 Source of operation command 3.2 Source of master frequency 3.3 Acceleration time 3.4 Deceleration time 3.5 Analogue input type 3.6 Digital output

Curricular Content Guide

1. Underpinning knowledge	1.1 Functions of Variable Frequency Drive (VFD) 1.2 Key components of VFD 1.3 Motor parameters and controlling parameters
2. Underpinning skills	2.1 Identifying common applications of VFDs in industrial automation 2.2 Installing VFD unit securely according to the manufacturer's guidelines 2.3 Installing and integrating controlling circuit for the VFD with the control system 2.4 Setting motor parameters accurately in the VFD 2.5 Configuring controlling parameters according to application requirements 2.6 Performing motor speed control using local or keypad control mode as per operational procedures 2.7 Executing motor control using terminal control mode or remote-control mode 2.8 Applying motor direction control to enable both forward and reverse operations safely 2.9 Monitoring and adjusting operational parameters to maintain desired motor performance
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns

	3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Personal protective equipment (PPE) 4.4 Hardware (PLC, VFD) 4.5 PLC programming software 4.6 VFD configuration tools 4.7 Drawings, specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 interpreted functions of variable frequency drive (VFD) 1.2 described key components of VFD 1.3 installed VFD unit securely according to the manufacturer's guidelines 1.4 installed and integrated controlling circuit for the VFD with the control system 1.5 configured controlling parameters according to application requirements 1.6 applied motor direction control to enable both forward and reverse operations safely
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DESIGN AND DEVELOP PLC BASED CONTROL SYSTEM	Nominal Duration: 55 hrs.	Unit Code: SICIP-LE-IA-08-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to design and develop PLC based control system. It specifically includes the tasks of analyzing control system requirements, designing PLC control system architecture, develop PLC program, simulating the PLC program, installing and configuring PLC hardware, and commissioning and debugging control system.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Analyze Control System Requirements	1.1 Control system objectives and functional requirements are identified according to <u>projects</u> . 1.2 System inputs, outputs, and interfaces are analyzed to determine operational needs. 1.3 <u>Hardware and software requirements</u> for the control system are specified and evaluated. 1.4 Constraints and limitations of the control system are identified and addressed in the analysis.
2. Design PLC Control System Architecture	2.1 Flow charts or algorithms for the PLC control system are developed according to project requirements. 2.2 Process steps and control sequences are mapped to ensure effective automation. 2.3 Control inputs, outputs, and feedback loops are incorporated into the system design. 2.4 Safety interlocks and error handling mechanisms are integrated into the control architecture. 2.5 Designed architecture is reviewed and validated for completeness and accuracy.
3. Develop PLC Program	3.1 <u>Appropriate programming language</u> are selected and applied according to project requirements. 3.2 PLC programs are developed based on designed algorithms and control sequences. 3.3 Input and output devices are mapped within the program to ensure control. 3.4 Program code is written, structured, and commented to maintain readability and facilitate troubleshooting.
4. Simulate the PLC Program	4.1 PLC program simulation tools are selected and configured according to system requirements. 4.2 PLC program is simulated to verify logic, sequence, and functional accuracy. 4.3 Simulation results are analyzed to identify errors, bugs,

	or inefficiencies in the program. 4.4 Corrective actions are implemented based on simulation feedback to improve program performance.
5. Install and Configure PLC Hardware	5.1 Installation site is inspected and prepared to ensure compliance with safety and operational standards. 5.2 <u>PLC hardware components</u> are identified and assembled. 5.3 PLC hardware is installed and connected according to project requirements. 5.4 Power and signal cables are connected correctly to PLC modules and peripheral devices. 5.5 Hardware configuration settings are set according to system requirements. 5.6 Functional tests are conducted to verify correct hardware installation.
6. Commission and Debug Control System	6.1 Program is tested on live hardware to validate operation under actual working conditions. 6.2 Modifications and debugging are performed as required. 6.3 System responses are monitored and evaluated. 6.4 Faults and malfunctions are identified, analyzed, and resolved systematically. 6.5 Commissioning procedures are documented and reported.

Range of Variables

Variable	Range
	May include but not limited to:
1. Projects	1.1 Conveyor belt operation 1.2 Auto door control 1.3 Motor forward and reverse operation 1.4 Motor star delta operation
2. Hardware and software requirements	2.1. Software 2.1.1 TIA Portal 2.1.2 GX Works 2.1.3 RSLOGIX 2.1.4 LSXG5000 2.1.5 WPLSoft 2.2. Hardware 2.1.1 PLC CPU 2.1.2 Input module 2.1.3 Output module 2.1.4 Communication module 2.1.5 Communication cable 2.1.6 Programming cable
3. Appropriate programming language	3.1 Ladder Logic 3.2 Function Block Diagram
4. PLC hardware components	4.1 CPU

	4.2 Power supply 4.3 Input/output modules 4.4 Communication modules
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Curricular Content Guide

1. Underpinning knowledge	1.1 Control system objectives and functional requirements (according to projects) 1.2 Constraints and limitations of the control system in the process of analysis 1.3 Appropriate programming languages 1.4 PLC program simulation tools 1.5 PLC hardware components
2. Underpinning skills	2.1 Specifying and evaluating hardware and software requirements for the control system 2.2 Developing flow charts or algorithms for the PLC control system according to project requirements 2.3 Incorporating control inputs, outputs and feedback loops into the system design 2.4 Designing PLC Control System Architecture 2.5 Selecting and applying appropriate programming language according to project requirements 2.6 Developing PLC programs based on designed algorithms and control sequences. 2.7 Simulating PLC program to verify logic, sequence and functional accuracy 2.8 Analyzing simulation results to identify errors, bugs or inefficiencies in the program and taking corrective actions thereon 2.9 Identifying and assembling PLC hardware components 2.10 Installing and connecting PLC hardware according to project requirements 2.11 Fixing hardware configuration settings according to system requirements 2.12 Testing program on live hardware to validate operation under actual working conditions 2.13 Identifying, analyzing and resolving faults and malfunctions systematically
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual)

	4.2 Computer/laptop/notebook 4.3 PLC hardware/ software 4.4 Appropriate programming language 4.5 Drawings, specifications and work instructions
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 interpreted appropriate programming languages 1.2 described PLC program simulation tools 1.3 explained PLC hardware components 1.4 developed flow charts or algorithms for the PLC control system according to project requirements 1.5 designed PLC Control System Architecture 1.6 applied appropriate programming language according to project requirements 1.7 developed PLC programs based on designed algorithms and control sequences 1.8 simulated PLC program to verify logic, sequence and functional accuracy 1.9 installed and connected PLC hardware according to project requirements 1.10 fixed hardware configuration settings according to system requirements 1.11 tested program on live hardware to validate operation under actual working conditions
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Industrial Automation
Number of Trainees:	25

Course-wise Training Space:

- Computer Lab : 400 sft
- Workshop : 700-800 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and facilities	Required facilities
1.	Computer/Laptops with PLC software	25
2.	HMI (Human-Machine Interface)	10
3.	PLC with required modules	10
4.	Limit Switch	20
5.	Proximity Sensor	20
6.	Photo Electric	20
7.	Float Sensor	10
8.	Pressure Sensor	05
9.	Temperature Sensor RTD	10
10.	Temperature Sensor Thermocouple	10
11.	Push button switch (green)	40
12.	Push button switch (red)	20
13.	Selector switch	20
14.	Emergency stop switch	20
15.	Indication lamp	50
16.	Magnetic contactor	100
17.	Relay	100
18.	Timer	100
19.	Counter	25
20.	Three phase induction motor	25

21.	VFDs-Variable Frequency Drives	10
22.	Power supply	10
23.	Automatic voltage stabilizer	01

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Industrial Automation, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.